PART 1 OF 5 · SOLUTIONS

Stochastic Calculus, Step by Step

Part 1: maths foundations.
Solutions to every exercise



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 1 OF 5 · SOLUTIONS

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Part 1: Maths foundations. Solutions

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Stochastic Calculus, Step by Step. Part 1: Maths foundations. Solutions.

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The course's numbers: $S_0 = 100$, $K = 105$, $r = 3\%$ (continuously compounded), $\sigma = 20\%$, $T = 1$ year. Made up for teaching; every random number is simulated with a fixed seed.

This volume holds the full solutions to the exercises and self-test questions in Stochastic Calculus, Step by Step, Part 1. Try each exercise before reading its solution.

Every number in the solutions is computed by a script that checks itself with assertions; the code solutions are shown in full.

This book is for education only. It is not financial, investment, legal or tax advice.

Contents

How to use the solutions 5

1. The Calculus You Need 6

2. Multivariable Calculus 12

3. Probability Basics 18

4. The Normal Distribution 23

5. Lognormal Prices and Jensen's Inequality 29

6. Convergence and Inequalities 34

How to use the solutions

This volume holds the full solutions to *Stochastic Calculus, Step by Step, Part 1: Maths foundations*: every exercise worked out, and the answers to the self-test questions at the end of each lesson. It is meant to be read **after** you have tried each one.

- **By-hand exercises** (the first six of each lesson) are worked in full. Each solution starts with **the idea** (the one thought that unlocks it), then goes through numbered steps with the reason for each, and ends with a check or an interpretation. If you are stuck, read only the idea and try again before reading the steps.
- **Code exercises** (the last two of each lesson) show a complete Python solution and exactly what it prints. Every number in this volume is produced by a script that checks itself with assertions, using the course's fixed random seeds, so your own run should match to the digits shown when you use the same seeds.
- **Self-test answers** are short: a sentence or two, and a number where there is one.

The course's numbers are the same throughout: $S_0 = 100$, $K = 105$, $r = 3\%$ (continuously compounded), $\sigma = 20\%$, $T = 1$ year.

If your answer differs, first check the usual suspects: a variance written where a standard deviation belongs (or the other way round), a missing factor of 2 on a cross term, and a log return confused with a simple return.

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SOLUTIONS TO LESSON 1

The Calculus You Need

Solutions to Exercises 1.1 to 1.8 and the Lesson 1 self-test.

Exercise 1.1

The idea. Both are functions of a function: use the chain rule, outer derivative times inner derivative.

$f(x) = e^{-x^2/2}$.

- Outer function e^u, with derivative e^u.
- Inner function $u = -x^2/2$, with derivative $u' = -\frac{1}{2} \cdot 2x = -x$ (power rule).
- Chain rule: outer derivative at the inner function, times inner derivative:

$$f'(x) = e^{-x^2/2} \cdot (-x) = -x\, e^{-x^2/2}.$$

$g(S) = \ln(Se^{rT})$. Two routes, and they must agree.

- *Chain rule.* Outer $\ln u$ with derivative $1/u$; inner $u = Se^{rT}$ with derivative e^{rT} (e^{rT} is a constant when S varies). So $g'(S) = \dfrac{1}{Se^{rT}} \cdot e^{rT} = \dfrac{1}{S}$.
- *Simplify first.* By $\ln(ab) = \ln a + \ln b$, $g(S) = \ln S + rT$. The constant rT has derivative 0, so $g'(S) = 1/S$.

At $S = 100$: $g'(100) = 0.01$. Simplifying first is often the faster route.

Exercise 1.2

The idea. Compute f, f' and f'' at the point $a = 1$, and put them into $f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$.

Step 1. Derivatives. Write $f(x) = x^{1/2}$ and use the power rule twice:

$$f'(x) = \tfrac{1}{2}x^{-1/2}, \qquad f''(x) = \tfrac{1}{2} \cdot (-\tfrac{1}{2})x^{-3/2} = -\tfrac{1}{4}x^{-3/2}.$$

At $x = 1$ every power of 1 is 1: $f(1) = 1$, $f'(1) = \frac{1}{2}$, $f''(1) = -\frac{1}{4}$.

Step 2. The polynomial. With $h = x - 1$:

$$\sqrt{x} \approx 1 + \tfrac{1}{2}(x-1) + \frac{-1/4}{2}(x-1)^2 = 1 + \tfrac{1}{2}(x-1) - \tfrac{1}{8}(x-1)^2.$$

Step 3. Evaluate at $x = 1.21$. Here $x - 1 = 0.21$, $\tfrac{1}{2}(0.21) = 0.105$ and $\tfrac{1}{8}(0.21)^2 = 0.0055125$, so the approximation is $1 + 0.105 - 0.0055125 = 1.0994875$.

Step 4. The error. Since $1.1^2 = 1.21$, the exact value is 1.1, and the error is -0.0005125.

Check with the remainder. The next term is $\frac{f'''(1)}{3!}(x-1)^3$, with $f'''(x) = \frac{3}{8}x^{-5/2}$, so it is $\frac{3/8}{6}(0.21)^3 = \frac{1}{16}(0.21)^3 = 0.0005788$. The error is close to the first term left out, as Taylor's theorem says.

Exercise 1.3

Step 1. Antiderivative. By the chain rule, $\frac{d}{dt}e^{-rt} = -r\,e^{-rt}$, so $\frac{d}{dt}\Big(-\frac{e^{-rt}}{r}\Big) = e^{-rt}$. An antiderivative is $-e^{-rt}/r$.

Step 2. Fundamental theorem.

$$\int_0^T e^{-rt}\,dt = \Big[-\frac{e^{-rt}}{r}\Big]_0^T = -\frac{e^{-rT}}{r} - \Big(-\frac{1}{r}\Big) = \frac{1 - e^{-rT}}{r}.$$

Step 3. Numbers. With $r = 3\%$, $T = 1$: 0.985149.

Interpretation. Receiving 1 a year, continuously, for a year is worth slightly less than 1 today, because the later payments are discounted more.

Check. As $r \to 0$ there is no discounting, so the answer should tend to T. By Taylor, $e^{-rT} \approx 1 - rT$, so $(1 - e^{-rT})/r \approx rT/r = T$. Correct.

Exercise 1.4

The idea. A power is hard to take a limit of; its logarithm is a product, which is easier. Take logarithms, use Taylor, then exponentiate back.

Step 1. Take logarithms. By $\ln(a^b) = b\ln a$,

$$\ln\Big(1 + \frac{r}{n}\Big)^n = n\ln\Big(1 + \frac{r}{n}\Big).$$

Step 2. Taylor. Put $x = r/n$ (small when n is large) into $\ln(1+x) = x - \tfrac{1}{2}x^2 + \dots$:

$$\ln\Big(1 + \frac{r}{n}\Big) = \frac{r}{n} - \frac{r^2}{2n^2} + \dots$$

Step 3. Multiply by n.

$$n \ln\left(1 + \frac{r}{n}\right) = r - \frac{r^2}{2n} + \dots$$

Every term after r has at least one power of n in the denominator, so the whole thing tends to r as $n \to \infty$.

Step 4. Exponentiate back. $(1 + r/n)^n = e^{n \ln(1+r/n)}$, and e^x is continuous, so $(1 + r/n)^n \to e^r$. $\square$

How fast. The gap $r - n\ln(1 + r/n)$ is about $r^2/(2n)$. With $r = 3\%$ it is 4.4e-04 for $n = 1$, 3.7e-05 for $n = 12$ and 1.2e-06 for $n = 365$, against $r^2/(2n) = 1.2\text{e-}06$ for daily compounding.

Exercise 1.5

The idea. The integrand is a product, $t \times e^{-rt}$. Differentiating t makes it simpler (it becomes 1), and e^{-rt} is easy to integrate. So use integration by parts with $u = t$ and $v' = e^{-rt}$.

Step 1. The pieces. $u = t$, so $u' = 1$. $v' = e^{-rt}$, so $v = -e^{-rt}/r$ (Exercise 1.3).

Step 2. The formula $\int_0^T uv'\,dt = [uv]_0^T - \int_0^T u'v\,dt$:

$$\int_0^T t\,e^{-rt}\,dt = \Big[-\frac{t\,e^{-rt}}{r}\Big]_0^T - \int_0^T \Big(-\frac{e^{-rt}}{r}\Big)dt = -\frac{T\,e^{-rT}}{r} + \frac{1}{r}\int_0^T e^{-rt}\,dt.$$

Step 3. Reuse Exercise 1.3 for the last integral:

$$\int_0^T t\,e^{-rt}\,dt = -\frac{T\,e^{-rT}}{r} + \frac{1 - e^{-rT}}{r^2}.$$

Step 4. Numbers. With $r = 3\%$, $T = 1$ the two terms are -32.3482 and 32.8383, so the integral is 0.490112. (The trapezoid rule gives 0.490112.)

Interpretation. Without discounting the answer would be $\int_0^1 t\,dt = \frac{1}{2}$; discounting reduces it slightly. Integrals like this one, payments weighted by their time, measure a bond's **duration**.

Notice how two large numbers nearly cancel. When you compute such a formula by hand, keep plenty of digits.

Exercise 1.6

Step 1. Derivatives. From the lesson, $P'(r) = -T\,e^{-rT}$ and $P''(r) = T^2 e^{-rT}$. With $T = 1$ and $r = 3\%$: $P = 0.970446$, $P' = -0.970446$ and $P'' = 0.970446$.

Step 2. First order. With $h = \Delta r = 0.01$:

$$\Delta P \approx P'(r)\,h = -0.970446 \times 0.01 = -0.009704.$$

Step 3. Second order. Add $\frac{1}{2}P''(r)h^2 = \frac{1}{2} \times 0.9704 \times 0.0001 = 0.000049$:

$$\Delta P \approx -0.009704 + 0.000049 = -0.009656.$$

Step 4. Exact. $\Delta P = e^{-0.04} - e^{-0.03} =$ -0.009656.

Compare. The first-order error is -4.8e-05; the second-order error is only 1.6e-07. The second-order term is positive because $P'' > 0$: the price falls *less* than the straight-line estimate, because the price curve bends upwards. That is the bond's **convexity**, and it always works in the holder's favour.

Exercise 1.7

Forward and central differences for the derivative of e^x at 0.

```
import numpy as np

rows = []
for k in range(1, 11):
    h = 10.0 ** -k
    fwd = (np.exp(h) - np.exp(0.0)) / h - 1.0          # the exact derivative at 0 is e^0 = 1
    cen = (np.exp(h) - np.exp(-h)) / (2 * h) - 1.0
    rows.append((h, abs(fwd), abs(cen)))
    print(f"h = {h:.0e}: forward error {abs(fwd):.1e}, central error {abs(cen):.1e}")
```

What it prints.

```
h = 1e-01: forward error 5.2e-02, central error 1.7e-03
h = 1e-02: forward error 5.0e-03, central error 1.7e-05
h = 1e-03: forward error 5.0e-04, central error 1.7e-07
h = 1e-04: forward error 5.0e-05, central error 1.7e-09
h = 1e-05: forward error 5.0e-06, central error 1.2e-11
h = 1e-06: forward error 5.0e-07, central error 2.7e-11
h = 1e-07: forward error 4.9e-08, central error 5.3e-10
h = 1e-08: forward error 6.1e-09, central error 5.3e-10
h = 1e-09: forward error 8.3e-08, central error 2.7e-08
h = 1e-10: forward error 8.3e-08, central error 8.3e-08
```

Why the errors fall. By Taylor, $e^h = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \ldots$, so

$$\frac{e^h - 1}{h} = 1 + \frac{h}{2} + \frac{h^2}{6} + \ldots$$

The forward difference is off by about $h/2$: this is its **truncation error**, from cutting the Taylor series short. For the central difference, subtract the series for e^{-h}:

$$e^h - e^{-h} = 2h + \frac{2h^3}{6} + \dots, \qquad \text{so} \qquad \frac{e^h - e^{-h}}{2h} = 1 + \frac{h^2}{6} + \dots$$

The even powers cancel, so its truncation error is about $h^2/6$, much smaller.

Why they rise again. A computer stores numbers to about 16 significant digits (the machine precision is $\epsilon \approx 2.2 \times 10^{-16}$). The numerator $e^h - 1$ subtracts two nearly equal numbers, so it carries an absolute **rounding error** of about ϵ. Dividing by a tiny h blows that up to about ϵ/h.

The best step. The total error is about $h/2 + \epsilon/h$. Its first part falls and its second part grows as h shrinks. They balance near $h \approx \sqrt{\epsilon} \approx 10^{-8}$ for the forward difference (best here at $h =$ 1.0e-08), and near $h \approx \epsilon^{1/3} \approx 10^{-5}$ for the central one (best at $h =$ 1.0e-05, error 1.2e-11). Smaller steps are not always better.

Exercise 1.8

Simple and log returns of the path 100, 110, 99, 108.9.

```
path = np.array([100.0, 110.0, 99.0, 108.9])
simple = path[1:] / path[:-1] - 1
logret = np.log(path[1:] / path[:-1])
print("simple returns:", simple.round(4), "sum", round(simple.sum(), 4), "whole period",
round(path[-1] / path[0] - 1, 4))
print("log returns:   ", logret.round(4), "sum", round(logret.sum(), 4), "whole period",
round(np.log(path[-1] / path[0]), 4))
```

What it prints.

```
simple returns: [ 0.1 -0.1  0.1] sum 0.1 whole period 0.089
log returns:    [ 0.0953 -0.1054  0.0953] sum 0.0853 whole period 0.0853
```

Simple returns. They are +10%, −10%, +10%, adding to 10%, but the share actually rose 0.0890, that is 8.9%. Simple returns **compound**: $(1.1)(0.9)(1.1) = 1.089$.

Log returns. They add up to 0.0853, which is exactly the whole period's log return $\ln(108.9/100)$, because

$$\ln\frac{S_1}{S_0} + \ln\frac{S_2}{S_1} + \ln\frac{S_3}{S_2} = \ln\left(\frac{S_1}{S_0}\cdot\frac{S_2}{S_1}\cdot\frac{S_3}{S_2}\right) = \ln\frac{S_3}{S_0}.$$

Log returns **add**.

Self-test

1. For a smooth path split into n steps of length $h = 1/n$, each second-order term is about $\frac{1}{2}f''h^2$, and n of them add up to about $\frac{1}{2}f''\,nh^2 = \frac{1}{2}f''h \to 0$: they can be dropped. A random step of size $\sigma\sqrt{\Delta t}$ has a square of size $\sigma^2\Delta t$. Over one day $\frac{1}{2}\sigma^2\Delta t =$ 7.9e-05, the same size as the drift $r\Delta t =$ 1.2e-04, and over a year the squares add up to σ^2, which does not vanish. The term must be kept.

2. $100e^{0.03} = 103.05$, against 103 for simple interest: a gap of 0.0455, the interest earned on interest.

3. $\ln(S_T/S_0) = 0.05$ means $S_T/S_0 = e^{0.05}$, so the simple return is $e^{0.05} - 1 = 5.1271\%$. It is a little more than 5%, as the lesson's rule "log return ≈ simple return − (simple return)^2/2" predicts.

SOLUTIONS TO LESSON 2

Multivariable Calculus

Solutions to Exercises 2.1 to 2.8 and the Lesson 2 self-test.

Exercise 2.1

The idea. For each partial derivative, freeze the other variable and treat it as a number.

$f(t, x) = \ln x + (r - \frac{1}{2}\sigma^2)(T - t)$.

- f_t. Freeze x: $\ln x$ is a constant, with derivative 0. In the second term, $(r - \frac{1}{2}\sigma^2)$ is a constant multiplier and $\frac{d}{dt}(T - t) = -1$. So

$$f_t = -(r - \tfrac{1}{2}\sigma^2),$$

which is -0.0100 with the course's numbers.

- f_x. Freeze t: the second term is a constant. So $f_x = 1/x$, which is 0.01 at $x = 100$.
- f_{xx}. Differentiate $1/x = x^{-1}$ again: $f_{xx} = -x^{-2} = -1/x^2$, which is -0.0001 at $x = 100$.

This f is the logarithm of a share price plus its expected drift; Lesson 16 applies Itô's lemma to exactly this kind of function, and needs exactly these three derivatives.

Exercise 2.2

Route 1: the chain rule. The pieces:

- $f_t = x^2$ (freeze x, differentiate t);
- $f_x = 2tx$ (freeze t, differentiate x^2);
- $x'(t) = ae^{at}$ (chain rule).

Then

$$\frac{d}{dt} f(t, x(t)) = f_t + f_x\, x'(t) = x(t)^2 + 2t\, x(t) \cdot ae^{at}.$$

Put in $x(t) = e^{at}$, so $x^2 = e^{2at}$ and $x \cdot e^{at} = e^{2at}$:

$$\frac{d}{dt} f(t, x(t)) = e^{2at} + 2at\, e^{2at} = e^{2at}(1 + 2at).$$

Route 2: substitute first. $f(t, x(t)) = t\, e^{2at}$, a function of t alone. The product rule gives $1 \cdot e^{2at} + t \cdot 2ae^{2at} = e^{2at}(1 + 2at)$: the same.

Numbers. With $a = 0.03$ at $t = 1$ both give 1.125547.

Exercise 2.3

Step 1. All the derivatives. For $f(x, y) = xy$: $f_x = y$, $f_y = x$, $f_{xx} = 0$, $f_{yy} = 0$, $f_{xy} = 1$, and every third derivative is 0. At $(1, 2)$: $f = 2$, $f_x = 2$, $f_y = 1$.

Step 2. The expansion. The second-order part is $\frac{1}{2}\big(f_{xx}\Delta x^2 + 2f_{xy}\Delta x\,\Delta y + f_{yy}\Delta y^2\big) = \frac{1}{2}(0 + 2\Delta x\,\Delta y + 0) = \Delta x\,\Delta y$. So

$$f(1 + \Delta x, 2 + \Delta y) \approx 2 + 2\Delta x + \Delta y + \Delta x\,\Delta y.$$

Step 3. It is exact. Multiply out directly: $(1 + \Delta x)(2 + \Delta y) = 2 + \Delta y + 2\Delta x + \Delta x\,\Delta y$. The same. (The remainder involves third derivatives, which are all zero.)

Step 4. Without the cross term the expansion is $2 + 2\Delta x + \Delta y$, so the error is exactly $\Delta x\,\Delta y$. With $\Delta x = 0.1$, $\Delta y = -0.2$: the true value is 1.98, the expansion without the cross term gives 2.00, and the difference is -0.02.

Interpretation. The cross term is the whole of the "interaction" between the two moves: the change you get only because both moved at once.

Exercise 2.4

The idea. Use the integrating factor e^{-kt}, as in the lesson.

Step 1. Rewrite the equation as $m' - km = c$ and multiply by e^{-kt}. By the product rule, the left side is a derivative:

$$\frac{d}{dt}\big(m\, e^{-kt}\big) = m'e^{-kt} - k\, m\, e^{-kt} = (m' - km)\, e^{-kt} = c\, e^{-kt}.$$

Step 2. Integrate from 0 to t. The left side gives $m(t)e^{-kt} - m(0)e^{0} = m(t)e^{-kt} - m_0$. The right side (Lesson 1, Exercise 1.3) gives

$$\int_0^t c\, e^{-ks}\, ds = \frac{c}{k}\big(1 - e^{-kt}\big).$$

Step 3. Solve for $m(t)$. Multiply by e^{kt}:

$$m(t) = m_0 e^{kt} + \frac{c}{k}\left(e^{kt} - 1\right) = -\frac{c}{k} + \left(m_0 + \frac{c}{k}\right)e^{kt}.$$

Check. At $t = 0$: $-c/k + m_0 + c/k = m_0$. And $m'(t) = k(m_0 + c/k)e^{kt} = k\big(m(t) + c/k\big) = k\,m(t) + c$. Both hold.

Numbers. With $k = 0.1$, $c = 5$, $m_0 = 100$: $c/k = 50$, so $m(1) = -50 + 150e^{0.1} = 115.7756$.

Interpretation. A bank account earning 10% continuously, with 5 a year paid in continuously.

Exercise 2.5

First derivatives. For $f(t, x) = t\,x^2 + e^{tx}$. The chain rule on e^{tx} brings down the derivative of the exponent tx: x when differentiating in t, and t when differentiating in x.

$$f_t = x^2 + x\,e^{tx}, \qquad f_x = 2tx + t\,e^{tx}.$$

Second derivatives in one variable.

$$f_{tt} = x \cdot x\,e^{tx} = x^2 e^{tx}, \qquad f_{xx} = 2t + t \cdot t\,e^{tx} = 2t + t^2 e^{tx}.$$

Mixed derivatives, both orders.

- f_{tx}: differentiate $f_t = x^2 + x\,e^{tx}$ in x. The product rule on $x\,e^{tx}$ gives $e^{tx} + x \cdot t\,e^{tx}$, so $f_{tx} = 2x + e^{tx} + tx\,e^{tx}$.
- f_{xt}: differentiate $f_x = 2tx + t\,e^{tx}$ in t. The product rule on $t\,e^{tx}$ gives $e^{tx} + t \cdot x\,e^{tx}$, so $f_{xt} = 2x + e^{tx} + tx\,e^{tx}$.

They are equal, as the symmetry theorem says.

Numbers. At $(1, 0.5)$, $e^{tx} = e^{0.5} = 1.648721$, so

$$f_{tx} = 2(0.5) + e^{0.5} + 0.5\,e^{0.5} = 1 + 1.5\,e^{0.5} = 3.473082.$$

A finite difference gives 3.473082.

Exercise 2.6

$f(t, x) = e^{x - t/2}$**.** By the chain rule, differentiating in t brings down $-\frac{1}{2}$ and differentiating in x brings down 1:

$$f_t = -\tfrac{1}{2}e^{x - t/2}, \qquad f_x = e^{x - t/2}, \qquad f_{xx} = e^{x - t/2}.$$

So $f_t + \frac{1}{2}f_{xx} = -\frac{1}{2}e^{x - t/2} + \frac{1}{2}e^{x - t/2} = 0$.

$g(t,x) = x^2 - t$. $g_t = -1$, $g_x = 2x$, $g_{xx} = 2$, so $g_t + \frac{1}{2}g_{xx} = -1 + 1 = 0$.

$h(t,x) = x^3 - c\,t\,x$. $h_t = -cx$, $h_x = 3x^2 - ct$, $h_{xx} = 6x$, so

$$h_t + \tfrac{1}{2}h_{xx} = -cx + 3x = (3-c)\,x.$$

This is zero for every x exactly when $c = 3$.

Why this matters. The equation $f_t + \frac{1}{2}f_{xx} = 0$ is the **heat equation** run backwards. Lesson 16 shows that when x is a Brownian motion, $f(t,x)$ has no drift exactly when f satisfies it: that is how $e^{W_t - t/2}$, $W_t^2 - t$ and $W_t^3 - 3tW_t$ turn out to be martingales.

Exercise 2.7

Halving both steps and watching the second-order error.

```
import math

r = 0.03
g = lambda t, x: math.exp(-r * t) * x**2
t0, x0 = 0.5, 100.0
gt, gx = -r * math.exp(-r * t0) * x0**2, 2 * math.exp(-r * t0) * x0
gtt, gtx, gxx = r**2 * math.exp(-r * t0) * x0**2, -2 * r * math.exp(-r * t0) * x0, 2 *
math.exp(-r * t0)
dt, dx = 0.2, 20.0
errors = []
for _ in range(5):
    taylor = gt * dt + gx * dx + 0.5 * (gtt * dt**2 + 2 * gtx * dt * dx + gxx * dx**2)
    errors.append(abs(g(t0 + dt, x0 + dx) - g(t0, x0) - taylor))
    dt, dx = dt / 2, dx / 2
print("errors:", [f"{e:.2e}" for e in errors])
print("ratios:", [round(a / b, 2) for a, b in zip(errors, errors[1:])])
```

What it prints.

```
errors: ['2.29e+00', '2.86e-01', '3.58e-02', '4.48e-03', '5.60e-04']
ratios: [7.99, 7.99, 8.0, 8.0]
```

Why 8. The error of the second-order expansion is the remainder R, of third order: at most a constant times $(|\Delta t| + |\Delta x|)^3$. Halving both steps halves $|\Delta t| + |\Delta x|$, and $(h/2)^3 = h^3/8$, so the error falls by a factor of about 8. (The starting steps are deliberately large so the errors stay well above the computer's rounding.)

Exercise 2.8

Euler's method for m′ = km.

```
k, m0, T = 0.10, 10_000.0, 1.0
exact = m0 * math.exp(k * T)
for n in (10, 100, 1000, 10_000):
    m = m0
    for _ in range(n):
        m += k * m * T / n
    print(f"n = {n:>6}: Euler {m:.4f}, error {m - exact:.4f}, error x n {(m - exact) *
n:.2f}")
```

What it prints.

```
n =     10: Euler 11046.2213, error -5.4879, error x n -54.88
n =    100: Euler 11051.1570, error -0.5522, error x n -55.22
n =   1000: Euler 11051.6539, error -0.0553, error x n -55.25
n =  10000: Euler 11051.7037, error -0.0055, error x n -55.26
```

What you see. The error times n settles near a constant, so the error is proportional to $1/n$.

Why. Each Euler step multiplies m by $(1 + k\,\Delta t) = (1 + k/n)$ (here $T = 1$), so after n steps Euler gives exactly $m_0(1 + k/n)^n$. Exercise 1.4 showed

$$n \ln\left(1 + \frac{k}{n}\right) = k - \frac{k^2}{2n} + \dots,$$

so

$$m_0\left(1 + \frac{k}{n}\right)^n \approx m_0 e^k e^{-k^2/(2n)} \approx m_0 e^k\left(1 - \frac{k^2}{2n}\right),$$

using $e^{-y} \approx 1 - y$ for small y. The error is about $-m_0 e^k k^2/(2n)$, and the error times n tends to $-m_0 e^k k^2/2 = $ -55.26.

Self-test

1. $\frac{1}{2} g_{xx} \Delta x^2$, at about 1.5637. A typical one-day move has Δx of size $\sigma x \sqrt{\Delta t}$, so Δx^2 is of size $\sigma^2 x^2 \Delta t$: proportional to Δt, like the time term. The other two second-order terms carry Δt^2 or $\Delta t\,\Delta x$, of size $\Delta t^{3/2}$, and are far smaller.

2. $m(t) = 105 e^{-0.03t}$, so $m(1) = 101.8968$: the strike discounted for a year, the present value of 105 paid in a year.

3. $f_x = 1$: the forward gains exactly what the share gains, so one share hedges it. $f_t = -rKe^{-r(T-t)}$ (the chain rule on $e^{-r(T-t)}$ brings down $+r$, and there is a minus sign in front), which is -3.0569 at $t = 0$. As time passes, the present value of the 105 you will pay rises towards 105, so with the share price unchanged the forward loses value.

SOLUTIONS TO LESSON 3

Probability Basics

Solutions to Exercises 3.1 to 3.8 and the Lesson 3 self-test.

Exercise 3.1

The share is 80, 100 or 125 with probabilities 0.25, 0.45, 0.30.

Mean. Weight each value by its probability:

$$E[S] = 80(0.25) + 100(0.45) + 125(0.30) = 20 + 45 + 37.5 = 102.50.$$

Variance, by the shortcut $\mathrm{Var}(S) = E[S^2] - E[S]^2$.

- Square each value and weight: $E[S^2] = 6400(0.25) + 10{,}000(0.45) + 15{,}625(0.30) = 1600 + 4500 + 4687.5 = 10787.50$.
- Subtract the squared mean, $102.5^2 = 10{,}506.25$: $\mathrm{Var}(S) = 281.25$.
- The standard deviation is $\sqrt{281.25} = 16.7705$.

Probability above the strike. Only the outcome 125 is above 105, so $P(S > 105) = P(S = 125) = 0.30$.

Call payoff. The payoffs are 0, 0 and $125 - 105 = 20$:

$$E[\max(S - 105, 0)] = 0(0.25) + 0(0.45) + 20(0.30) = 6.00.$$

Exercise 3.2

Step 1. The density. Uniform on $[80, 130]$ means a flat density over an interval of length 50. Its area must be 1, so its height is $1/50$.

Step 2. Where the payoff is non-zero. $\max(x - K, 0)$ is 0 for $x \le K$ and $x - K$ for $x > K$. So only the part of the interval above K counts:

$$E[\max(X - K, 0)] = \int_K^{130} (x - K)\,\frac{1}{50}\,dx.$$

Step 3. Integrate. An antiderivative of $x - K$ is $(x - K)^2/2$ (check by differentiating):

$$\frac{1}{50}\left[\frac{(x-K)^2}{2}\right]_K^{130} = \frac{1}{50}\cdot\frac{(130-K)^2}{2} = \frac{(130-K)^2}{100}.$$

Step 4. Evaluate. At $K = 105$: $25^2/100 = 6.25$.

Check. At $K = 80$ the call is always exercised, so its payoff is just $X - 80$ and its average is $E[X] - 80 = 105 - 80 = 25$. The formula gives $50^2/100 = 25$. Correct.

Exercise 3.3

Let $\mu = E[X]$. By linearity, $E[2X + 50] = 2\mu + 50$.

Variance. Subtract the mean inside the definition; the 50s cancel:

$$\begin{aligned}\operatorname{Var}(2X+50) &= E\big[(2X + 50 - 2\mu - 50)^2\big] = E\big[(2(X-\mu))^2\big] \\ &= 4\,E\big[(X-\mu)^2\big] = 4\operatorname{Var}(X).\end{aligned}$$

Covariance. Same idea:

$$\begin{aligned}\operatorname{Cov}(X, 2X+50) &= E\big[(X-\mu)(2X + 50 - 2\mu - 50)\big] \\ &= 2\,E\big[(X-\mu)^2\big] = 2\operatorname{Var}(X).\end{aligned}$$

Numbers. For the triangular price, $\operatorname{Var}(X) = 205.5556$, so the two answers are 822.2222 and 411.1111.

Interpretation. The constant 50 changes neither: adding cash moves the mean, not the spread.

Exercise 3.4

Covariance.

- $E[Z] = (-1)\frac{1}{3} + 0\cdot\frac{1}{3} + 1\cdot\frac{1}{3} = 0$.
- $ZY = Z\cdot Z^2 = Z^3$, which takes the values -1, 0, 1, so $E[ZY] = 0$.
- So $\operatorname{Cov}(Z, Y) = E[ZY] - E[Z]E[Y] = 0 - 0 \times E[Y] = 0$.

Not independent. Independence would require, for example, $P(Y = 1 \text{ and } Z = 1) = P(Y = 1)\,P(Z = 1)$.

- $Y = 1$ when $Z = 1$ or $Z = -1$, so $P(Y = 1) = 0.6667$.
- $P(Z = 1) = \frac{1}{3}$, so the product is $\frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$.
- But $Z = 1$ forces $Y = 1$, so $P(Y = 1 \text{ and } Z = 1) = P(Z = 1) = \frac{1}{3} \neq \frac{2}{9}$.

Knowing Z changes the odds for Y, which independence forbids. Zero covariance only rules out a *straight-line* relation, and $Y = Z^2$ is not a straight line.

Exercise 3.5

Left of the peak, $70 \le x \le 100$. Integrate the density from 70 to x, substituting $u = s - 70$:

$$F(x) = \int_{70}^{x} \frac{s-70}{1050}\,ds = \frac{1}{1050}\int_{0}^{x-70} u\,du = \frac{(x-70)^2}{2100}.$$

At the peak, $F(100) = 900/2100 = 3/7 = 0.4286$: the area of the left triangle.

Right of the peak, $100 \le x \le 140$. It is easier to compute the area to the *right* of x and use the complement rule. Substituting $u = 140 - s$:

$$P(X > x) = \int_{x}^{140} \frac{140-s}{1400}\,ds = \frac{(140-x)^2}{2800}, \qquad \text{so} \qquad F(x) = 1 - \frac{(140-x)^2}{2800}.$$

Checks. At $x = 100$ this gives $1 - 1600/2800 = 1 - 4/7 = 3/7$, matching the left formula. At $x = 140$ it gives $1 - 0 = 1$.

The band. $P(90 < X < 110) = F(110) - F(90)$, with $F(90) = 20^2/2100 = 0.1905$ (left formula) and $F(110) = 1 - 30^2/2800 = 0.6786$ (right formula):

$$P(90 < X < 110) = 0.6786 - 0.1905 = 0.4881.$$

Exercise 3.6

Step 1. Covariance from correlation. By definition $\rho = \mathrm{Cov}(X,Y)/(\mathrm{sd}\,X \cdot \mathrm{sd}\,Y)$, so $\mathrm{Cov}(X,Y) = \rho \times 20 \times 20 = 400\rho$.

Step 2. Variance of the sum.

$$\mathrm{Var}(X+Y) = 400 + 400 + 2 \times 400\rho = 800(1+\rho).$$

Step 3. The four cases.

CORRELATION	VARIANCE OF THE SUM	STANDARD DEVIATION
1	1600	40.00
0.5	1200	34.64
0	800	28.28
−1	0	0.00

Interpretation. Only when the two move in perfect lockstep ($\rho = 1$) do the standard deviations add, to 40. With any lower correlation the risk of holding both is less than the sum of the two risks: that is **diversification**. At $\rho = -1$ the moves cancel exactly and the

sum is not random at all, the idea behind **hedging**.

Exercise 3.7

Inverse-transform sampling from the triangular density.

```
import numpy as np

a, c, b, K = 70.0, 100.0, 140.0, 105.0
rng = np.random.default_rng(1)
u = rng.random(1_000_000)
split = (c - a) / (b - a)                          # P(X < c), the area left of the peak
X = np.where(u < split, a + np.sqrt(u * (b - a) * (c - a)), b - np.sqrt((1 - u) * (b - a) *
(b - c)))
exact = {"mean": (a + b + c) / 3, "var": (a*a + b*b + c*c - a*b - a*c - b*c) / 18,
         "call": (b - K) ** 3 / (3 * (b - a) * (b - c))}
sim = {"mean": X.mean(), "var": X.var(), "call": np.maximum(X - K, 0).mean()}
for key in exact:
    print(f"{key}: simulated {sim[key]:.4f}, exact {exact[key]:.4f}")
```

What it prints.

```
mean: simulated 103.3331, exact 103.3333
var: simulated 205.6059, exact 205.5556
call: simulated 5.1060, exact 5.1042
```

Why the inverse transform works. Let U be uniform on $[0, 1]$, so $P(U \le u) = u$. If F is the distribution function of X, set $X' = F^{-1}(U)$. Because F is increasing,

$$P(X' \le x) = P\big(F^{-1}(U) \le x\big) = P\big(U \le F(x)\big) = F(x),$$

so X' has exactly the distribution of X.

For the triangle. Exercise 3.5 gave F. Solve $F(x) = u$ on each side:

- if $u \le 3/7$: $(x - 70)^2/2100 = u$, so $x = 70 + \sqrt{2100\,u}$;
- if $u > 3/7$: $1 - (140 - x)^2/2800 = u$, so $x = 140 - \sqrt{2800\,(1 - u)}$.

These are the two square-root formulas in the code ($2100 = 70 \times 30$ and $2800 = 70 \times 40$). All three simulated values are within about 0.1% of the exact ones.

Exercise 3.8

A variance identity, checked on a million correlated pairs.

```
rng = np.random.default_rng(2)
Xs = rng.normal(100, 20, 1_000_000)
Ys = 0.6 * Xs + rng.normal(40, 15, Xs.size)       # moves with X, plus its own noise
lhs = (Xs - Ys).var()
rhs = Xs.var() + Ys.var() - 2 * np.cov(Xs, Ys, ddof=0)[0, 1]
print(f"Var(X - Y) = {lhs:.3f}; Var X + Var Y - 2 Cov = {rhs:.3f}")
```

What it prints.

```
Var(X - Y) = 288.948; Var X + Var Y - 2 Cov = 288.948
```

Why both sides agree exactly. The identity is pure algebra, so it holds for the sample averages too, not just in theory (with `ddof=0`, the sample variance and covariance are averages of squares and products).

The theoretical value. $X \sim N(100, 20^2)$ and $Y = 0.6X + \varepsilon$ with $\varepsilon \sim N(40, 15^2)$ independent of X. Then $X - Y = 0.4X - \varepsilon$, and since X and ε are independent their variances add:

$$\mathrm{Var}(X - Y) = 0.4^2 \times 400 + 225 = 289.00,$$

while the simulation gives 288.95.

Self-test

1. $E[\max(X - 105, 0)] = \quad 5.1042$, while $\max(E[X] - 105, 0) = \max(103.33 - 105, 0) = 0.00$. They differ because the payoff is not a straight line: averaging the price first throws away the outcomes above 105 that make the call worth something. (Lesson 5: Jensen's inequality.)

2. Not necessarily. Zero covariance means no straight-line relation; U and U^2 (or Z and Z^2 in Exercise 3.4) have zero covariance and are completely dependent.

3. For independent variables the covariance is 0, and $\mathrm{Var}(-Y) = (-1)^2\mathrm{Var}(Y)$, so

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = 9 + 16 = 25,$$

and the standard deviation is 5: not $4 - 3 = 1$ or $3 + 4 = 7$. The minus sign does not reduce the spread.

SOLUTIONS TO LESSON 4

The Normal Distribution

Solutions to Exercises 4.1 to 4.8 and the Lesson 4 self-test.

Exercise 4.1

The idea. Every question about a normal X becomes a question about Z by standardising: subtract the mean, divide by the standard deviation.

Step 1. Read off the parameters. $X \sim N(100, 20^2)$ has mean $\mu = 100$ and variance 20^2, so standard deviation $s = 20$. Then $Z = (X - 100)/20$ is standard normal.

Step 2. $P(X > 105)$**.** Subtract 100 and divide by 20 on both sides of $X > 105$:

$$X > 105 \iff \frac{X - 100}{20} > \frac{105 - 100}{20} = 0.25.$$

So $P(X > 105) = P(Z > 0.25) = 1 - N(0.25) = 0.4013$.

Step 3. $P(90 < X < 110)$**.** Standardise both ends: $(90 - 100)/20 = -0.5$ and $(110 - 100)/20 = 0.5$, so

$$P(90 < X < 110) = P(-0.5 < Z < 0.5) = N(0.5) - N(-0.5).$$

Step 4. Use symmetry. $N(-0.5) = 1 - N(0.5)$, so the answer is $N(0.5) - (1 - N(0.5)) = 2N(0.5) - 1$. With $N(0.5) = 0.6915$, this is 0.3829.

Interpretation. About 38% of outcomes fall within half a standard deviation of the mean.

Exercise 4.2

The idea. The moments are hidden in the coefficients of the power series of M: the coefficient of u^k is $E[Z^k]/k!$.

Step 1. Expand M**.** Put $x = u^2/2$ in $e^x = \sum_j x^j/j!$:

$$e^{u^2/2} = \sum_{j \geq 0} \frac{(u^2/2)^j}{j!} = 1 + \frac{u^2}{2} + \frac{u^4}{2^2 \cdot 2!} + \frac{u^6}{2^3 \cdot 3!} + \cdots$$

Step 2. Find the u^6 coefficient. It comes from $j = 3$: $\dfrac{1}{2^3 \cdot 3!} = \dfrac{1}{8 \times 6} = \dfrac{1}{48}$.

Step 3. Match. In $M(u) = \sum_k E[Z^k]u^k/k!$ the coefficient of u^6 is $E[Z^6]/6!$. So

$$\frac{E[Z^6]}{720} = \frac{1}{48} \quad \Longrightarrow \quad E[Z^6] = \frac{720}{48} = 15.$$

Check. The pattern $E[Z^{2j}] = 1 \times 3 \times 5 \times \cdots \times (2j-1)$ gives $1 \times 3 \times 5 = 15$. The sequence of even moments is 1, 3, 15, 105, ...

Exercise 4.3

The idea. Use the two properties of generating functions: independent sums become products, and the generating function identifies the distribution.

Step 1. Each $X_i \sim N(\mu, s^2)$ has $M_{X_i}(u) = e^{u\mu + \frac{1}{2}u^2 s^2}$.

Step 2. The X_i are independent, so the expectation of the product $e^{uX_1} \cdots e^{uX_n}$ is the product of the expectations:

$$M_{X_1 + \cdots + X_n}(u) = E\left[e^{uX_1} \cdots e^{uX_n}\right] = \prod_{i=1}^{n} e^{u\mu + \frac{1}{2}u^2 s^2}.$$

Step 3. A product of n equal exponentials adds the exponent n times:

$$\prod_{i=1}^{n} e^{u\mu + \frac{1}{2}u^2 s^2} = e^{n(u\mu + \frac{1}{2}u^2 s^2)} = e^{u(n\mu) + \frac{1}{2}u^2 (ns^2)}.$$

Step 4. That is the generating function of $N(n\mu, ns^2)$ (mean $n\mu$, variance ns^2). By uniqueness, $X_1 + \cdots + X_n \sim N(n\mu, ns^2)$. $\square$

Interpretation. The mean grows like n but the standard deviation, $s\sqrt{n}$, only like $\sqrt{n}$.

Exercise 4.4

The idea. Turn the expectation into an integral over $z > a$, complete the square so the integrand becomes a shifted bell, and substitute.

Step 1. Complete the square. As in the lesson,

$$uz - \tfrac{1}{2}z^2 = -\tfrac{1}{2}(z-u)^2 + \tfrac{1}{2}u^2,$$

so, exponentiating both sides and dividing by $\sqrt{2\pi}$,

$$e^{uz}\varphi(z) = e^{u^2/2}\varphi(z-u).$$

Step 2. Write the expectation as an integral. The indicator keeps only $z > a$:

$$E\big[e^{uZ}\mathbf{1}\{Z > a\}\big] = \int_a^\infty e^{uz}\varphi(z)\,dz = e^{u^2/2}\int_a^\infty \varphi(z-u)\,dz.$$

Step 3. Substitute $w = z - u$. Then $dw = dz$, and the lower limit $z = a$ becomes $w = a - u$:

$$e^{u^2/2}\int_{a-u}^\infty \varphi(w)\,dw = e^{u^2/2}\,P(Z > a-u) = e^{u^2/2}\big(1 - N(a-u)\big).$$

Step 4. Symmetry. $1 - N(a-u) = N(u-a)$, which gives $e^{u^2/2}N(u-a)$. □

Numbers. With $u = 0.2$, $a = 0$: $e^{0.02}N(0.2) = 1.0202 \times 0.5793 = 0.5910$.

Check. Without the indicator the answer would be $e^{0.02} = 1.0202$. The indicator keeps a bit more than half of it, because $e^{0.2Z}$ is larger on the outcomes $Z > 0$.

Exercise 4.5

Step 1. The generating function. X takes two values, each with probability $\frac{1}{2}$:

$$M_X(u) = \tfrac{1}{2}e^u + \tfrac{1}{2}e^{-u} = \frac{e^u + e^{-u}}{2}.$$

Step 2. Expand. Write out both exponential series:

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \dots, \qquad e^{-u} = 1 - u + \frac{u^2}{2!} - \frac{u^3}{3!} + \frac{u^4}{4!} - \dots$$

Adding them, the odd powers cancel and the even powers double; halving gives

$$M_X(u) = 1 + \frac{u^2}{2!} + \frac{u^4}{4!} + \frac{u^6}{6!} + \dots$$

Step 3. Match coefficients. The coefficient of u^2 is $E[X^2]/2! = 1/2!$, so $E[X^2] = 1$. The coefficient of u^4 is $E[X^4]/4! = 1/4!$, so $E[X^4] = 1$. (Directly: $X^2 = 1$ and $X^4 = 1$ always.)

Step 4. Compare. X and Z have the same mean (0) and the same variance (1), but $E[X^4] = 1$ while $E[Z^4] = 3$. The fourth power magnifies large values: X never goes beyond 1 in size, while Z sometimes takes values like 2 or 3, whose fourth powers are 16 and 81. So **the normal has heavier tails than the coin**. A larger fourth moment, for the same variance, means more frequent extreme outcomes.

Exercise 4.6

The idea. Do the same completing-the-square calculation as for Z, but with μ and s in the exponent.

Step 1. Write the integral. The density of $N(\mu, s^2)$ is $\frac{1}{s\sqrt{2\pi}} e^{-(x-\mu)^2/(2s^2)}$, so

$$M_X(u) = \frac{1}{s\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(ux - \frac{(x-\mu)^2}{2s^2}\right) dx.$$

Step 2. Complete the square in the exponent. Put everything over $2s^2$:

$$ux - \frac{(x-\mu)^2}{2s^2} = -\frac{(x-\mu)^2 - 2s^2ux}{2s^2}.$$

Expand the numerator: $x^2 - 2\mu x + \mu^2 - 2s^2ux = x^2 - 2(\mu + s^2u)x + \mu^2$. Add and subtract $(\mu + s^2u)^2$:

$$\begin{aligned} x^2 - 2(\mu + s^2u)x + \mu^2 &= \big(x - (\mu + s^2u)\big)^2 - (\mu + s^2u)^2 + \mu^2 \\ &= \big(x - (\mu + s^2u)\big)^2 - 2\mu s^2u - s^4u^2. \end{aligned}$$

Divide by $-2s^2$:

$$ux - \frac{(x-\mu)^2}{2s^2} = -\frac{\big(x - (\mu + s^2u)\big)^2}{2s^2} + u\mu + \tfrac{1}{2}u^2s^2.$$

Step 3. Take the constant outside. The last two terms do not contain x:

$$M_X(u) = e^{u\mu + \frac{1}{2}u^2s^2} \int_{-\infty}^{\infty} \frac{1}{s\sqrt{2\pi}} \exp\Big(-\frac{\big(x - (\mu + s^2u)\big)^2}{2s^2}\Big)\, dx.$$

Step 4. The integrand is the density of $N(\mu + s^2u,\ s^2)$, so the integral is 1, and

$$M_X(u) = e^{u\mu + \frac{1}{2}u^2s^2}, \ \square$$

the same as the lesson's answer by $X = \mu + sZ$. Notice the tilted density is again normal, with the same variance and the mean moved by s^2u.

Exercise 4.7

Simulated E[Z^4] and E|Z| at three sample sizes.

```
import numpy as np

rng = np.random.default_rng(3)
for n in (10**3, 10**5, 10**7):
    Z = rng.standard_normal(n)
    z4, az = (Z**4).mean(), np.abs(Z).mean()
    se4, seaz = (Z**4).std() / np.sqrt(n), np.abs(Z).std() / np.sqrt(n)
    print(f"n = {n:>8}: E[Z^4] ~ {z4:.4f} (se {se4:.4f}), E|Z| ~ {az:.4f} (se {seaz:.4f})")
print("exact: 3 and", round(np.sqrt(2 / np.pi), 4))
```

What it prints.

```
n =     1000: E[Z^4] ~ 3.2184 (se 0.3320), E|Z| ~ 0.7984 (se 0.0194)
n =   100000: E[Z^4] ~ 2.9751 (se 0.0306), E|Z| ~ 0.7977 (se 0.0019)
n = 10000000: E[Z^4] ~ 2.9959 (se 0.0031), E|Z| ~ 0.7974 (se 0.0002)
exact: 3 and 0.7979
```

What you see. $E|Z|$ settles much faster than $E[Z^4]$.

Why. A simulation estimate is an average of n independent draws, and Lesson 6 shows that its standard error (its typical distance from the truth) is

$$\frac{\text{standard deviation of one draw}}{\sqrt{n}}.$$

- For $|Z|$: $\mathrm{Var}|Z| = E[Z^2] - (E|Z|)^2 = 1 - 2/\pi$, so one draw has standard deviation $\sqrt{1 - 2/\pi} = 0.6028$.
- For Z^4: $\mathrm{Var}(Z^4) = E[Z^8] - E[Z^4]^2 = 105 - 9 = 96$ (using $E[Z^8] = 1 \times 3 \times 5 \times 7 = 105$), so one draw has standard deviation $\sqrt{96} = 9.80$.

That is about 16 times larger, because Z^4 is dominated by rare large draws: a single $Z = 4$ contributes 256. Both errors shrink like $1/\sqrt{n}$; one simply starts much bigger.

Exercise 4.8

The completed square as a price, checked by simulation.

```
import math

S0, K, r, sigma, T = 100.0, 105.0, 0.03, 0.20, 1.0
N = lambda x: 0.5 * (1 + math.erf(x / math.sqrt(2)))
Z = np.random.default_rng(4).standard_normal(4_000_000)
ST = S0 * np.exp((r - 0.5 * sigma**2) * T + sigma * math.sqrt(T) * Z)
x = math.exp(-r * T) * ST * (ST > K)
d_plus = (math.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
print(f"simulated {x.mean():.4f} +/- {x.std() / math.sqrt(x.size):.4f}; S0 N(d+) = {S0 *
N(d_plus):.4f}")
```

What it prints.

```
simulated 50.2348 +/- 0.0297; S0 N(d+) = 50.2413
```

What the code does. It simulates $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$ four million times, sets each price to 0 when $S_T \le K$ (the `(ST > K)` factor is the indicator), discounts, and averages.

The result. With $d_+ = 0.006049$, the formula gives $S_0 N(d_+) = 50.2413$. The simulation gives 50.2348 with a standard error of 0.0297: the two agree to within one standard error. This is the first of the two terms of the Black-Scholes call price.

Self-test

1. Use $\mathrm{Var}(Y) = E[Y^2] - E[Y]^2$ with $Y = Z^2$:

$$\mathrm{Var}(Z^2) = E[Z^4] - E[Z^2]^2 = 3 - 1^2 = 2.$$

Lesson 6 uses it: a sum of n independent squared steps, each of variance T/n, has variance $n \times (T/n)^2 \times 2 = 2T^2/n$, which goes to 0.

2. It is the generating function at $u = -0.5$:

$$E[e^{-0.5Z}] = e^{(-0.5)^2/2} = e^{0.125} = 1.1331.$$

The sign of u doesn't matter, because u enters squared: the exponential is convex either way, and the average is pushed above $e^0 = 1$.

3. The standard normal density is symmetric about 0, so the area to the right of x equals the area to the left of $-x$: $1 - N(x) = N(-x)$. (The lesson proves it by substituting $w = -z$.) The lesson used it to write $P(S_T > K) = 1 - N(z_K) = N(-z_K)$, and in completing the square, $1 - N(a - u) = N(u - a)$. Every Black-Scholes formula is written with it.

SOLUTIONS TO LESSON 5

Lognormal Prices and Jensen's Inequality

Solutions to Exercises 5.1 to 5.8 and the Lesson 5 self-test.

Exercise 5.1

The idea. Every moment of a lognormal is a value of the normal's generating function.

Write $S_T = S_0 e^{aT+bZ}$ with $a = r - \frac{1}{2}\sigma^2$ and $b = \sigma\sqrt{T}$.

Mean ($u = b$). Take the constant $S_0 e^{aT}$ outside the expectation:

$$E[S_T] = S_0 e^{aT} E[e^{bZ}] = S_0 e^{aT+b^2/2} = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\frac{1}{2}\sigma^2 T} = S_0 e^{rT} = 103.0455.$$

Second moment ($u = 2b$). First square: $S_T^2 = S_0^2 e^{2aT+2bZ}$. Then

$$E[S_T^2] = S_0^2 e^{2aT} E[e^{2bZ}] = S_0^2 e^{2aT+(2b)^2/2} = S_0^2 e^{2aT+2b^2}.$$

With $2aT + 2b^2 = (2r - \sigma^2)T + 2\sigma^2 T = (2r + \sigma^2)T$, this is $S_0^2 e^{(2r+\sigma^2)T} = 11051.71$.

Variance.

$$\mathrm{Var}(S_T) = E[S_T^2] - E[S_T]^2 = S_0^2 e^{(2r+\sigma^2)T} - S_0^2 e^{2rT} = S_0^2 e^{2rT}\left(e^{\sigma^2 T} - 1\right) = 433.34,$$

so the standard deviation is 20.8169.

Exercise 5.2

Step 1. Move to logarithms. For any $M > 0$, because ln is increasing,

$$P(S_T \le M) = P(\ln S_T \le \ln M).$$

So M is the median of S_T exactly when $\ln M$ is the median of $\ln S_T$.

Step 2. The median of a normal. $\ln S_T$ is normal with mean $\ln S_0 + (r - \frac{1}{2}\sigma^2)T$. A normal density is symmetric about its mean, so half the probability is on each side: its median is its mean.

Step 3. Exponentiate.

$$\text{median}(S_T) = \exp\left(\ln S_0 + (r - \tfrac{1}{2}\sigma^2)T\right) = S_0 e^{(r-\frac{1}{2}\sigma^2)T} = 101.0050.$$

Why below the mean. The ratio median / mean is $e^{-\frac{1}{2}\sigma^2 T} < 1$. The lognormal is skewed right: a price can rise without limit but cannot fall below 0, so the occasional large rise pulls the average above the typical outcome. (The exponential is convex: Jensen.)

Exercise 5.3

$f(x) = x^2$ has $f'' = 2 > 0$, so it is convex, and Jensen gives

$$E[X^2] \geq \big(E[X]\big)^2, \qquad \text{that is,} \qquad \text{Var}(X) = E[X^2] - E[X]^2 \geq 0.$$

Variances are never negative. Equality holds only when X is constant, the only random variable with no spread. (Of course this also follows from the definition, an average of squares; Jensen gives a second view.)

Exercise 5.4

The idea. The peak of the density is where its derivative is 0. It is easier to maximise $\ln f$: the logarithm is increasing, so f and $\ln f$ peak at the same place.

Step 1. Take logarithms of the density. With $c = \ln S_0 + m$ and $m = (r - \frac{1}{2}\sigma^2)T$, $v = \sigma\sqrt{T}$:

$$\ln f(s) = -\ln s - \ln\left(v\sqrt{2\pi}\right) - \frac{(\ln s - c)^2}{2v^2}.$$

Step 2. Differentiate in s. The middle term is a constant. By the chain rule, $\frac{d}{ds}(\ln s - c)^2 = 2(\ln s - c)\cdot\frac{1}{s}$:

$$\frac{d}{ds}\ln f(s) = -\frac{1}{s} - \frac{\ln s - c}{v^2\, s}.$$

Step 3. Set it to zero. Multiply by $-s\,v^2$:

$$v^2 + \ln s - c = 0 \quad\Longrightarrow\quad \ln s = c - v^2 = \ln S_0 + (r - \tfrac{1}{2}\sigma^2)T - \sigma^2 T.$$

Step 4. Exponentiate.

$$\text{mode}(S_T) = S_0 e^{(r-\frac{3}{2}\sigma^2)T} = 97.0446.$$

(The exponent is -0.03.) The derivative is positive to the left of this point and negative to the right, so it is a maximum. Together with Exercise 5.2: mode < median < mean.

Exercise 5.5

Lower bound from Jensen. The put payoff is convex, so

$$E[\max(K - S_T, 0)] \geq \max\left(K - E[S_T], 0\right) = \max(105 - 103.0455, 0) = 1.9545.$$

Exact value.

Step 1. A payoff identity. For any s, $\max(K - s, 0) = \max(s - K, 0) - s + K$. (Check both cases. If $s > K$: left side 0, right side $(s - K) - s + K = 0$. If $s \leq K$: left side $K - s$, right side $0 - s + K$.)

Step 2. Take expectations (linearity):

$$E[\max(K - S_T, 0)] = E[\max(S_T - K, 0)] - E[S_T] + K.$$

Step 3. The call. Lesson 4 showed $e^{-rT}E[S_T \mathbf{1}\{S_T > K\}] = S_0 N(d_+)$ and $P(S_T > K) = N(d_-)$. Since $\max(S_T - K, 0) = S_T \mathbf{1}\{S_T > K\} - K\,\mathbf{1}\{S_T > K\}$,

$$E[\max(S_T - K, 0)] = S_0 e^{rT} N(d_+) - K\,N(d_-).$$

With $d_+ = 0.006049$, $N(d_+) = 0.502413$, d_- = -0.193951 and $N(d_-) = 0.423107$:

$$E[\max(S_T - K, 0)] = 103.0455 \times 0.502413 - 105 \times 0.423107 = 7.3451.$$

(The notebook's simulation gives 7.3455.)

Step 4. The put.

$$E[\max(K - S_T, 0)] = 7.3451 - 103.0455 + 105 = 9.2997.$$

Compare. The exact value, 9.2997, is well above the Jensen bound 1.9545: the bound only uses the average price, and the put gains a lot from the spread around it. (Discounting both by e^{-rT} gives the Black-Scholes put price; Step 1 is **put-call parity**.)

Exercise 5.6

Step 1. The sum's mean and variance. By linearity, the mean is $252 \times 0.01/252 = 0.01$. The days are independent, so variances add: $252 \times (0.2/\sqrt{252})^2 = 0.2^2$, a standard deviation of 0.2.

Step 2. The central limit theorem. A sum of 252 independent, identically distributed pieces is approximately normal, so the year's log return L is approximately $N(0.01, 0.2^2)$, whatever the daily distribution.

Step 3. Standardise.

$$P(L > 0.2) \approx P\Big(Z > \frac{0.2 - 0.01}{0.2}\Big) = P(Z > 0.95) = 1 - N(0.95) = 0.1711.$$

About a 17% chance of a year with a log return above 20%. The approximation is excellent near the middle of the distribution; far out in the tails it depends more on the daily distribution (Exercise 5.8).

Exercise 5.7

The same draws, with and without the $-\sigma^2/2$.

```
import math

import numpy as np

S0, r, T = 100.0, 0.03, 1.0
Z = np.random.default_rng(5).standard_normal(2_000_000)
for sigma in (0.20, 0.40):
    right = S0 * np.exp((r - 0.5 * sigma**2) * T + sigma * math.sqrt(T) * Z)
    naive = S0 * np.exp(r * T + sigma * math.sqrt(T) * Z)
    print(f"sigma {sigma:.0%}: with -sigma^2/2 {right.mean():.2f}, without
{naive.mean():.2f}; "
          f"S0 e^(rT) = {S0 * math.exp(r * T):.2f}")
```

What it prints.

```
sigma 20%: with -sigma^2/2 103.06, without 105.14; S0 e^(rT) = 103.05
sigma 40%: with -sigma^2/2 103.07, without 111.66; S0 e^(rT) = 103.05
```

What you see. With the correction, the average is $S_0 e^{rT}$ whatever σ is. Without it, the average is $S_0 e^{rT} e^{\frac{1}{2}\sigma^2 T}$: too big by about 2% at $\sigma = 20\%$ ($e^{0.02}$) and by more than 8% at $\sigma = 40\%$ ($e^{0.08}$).

Why. Jensen's gap is $e^{\frac{1}{2}\sigma^2 T}$, so it grows with the **square** of the volatility: doubling σ quadruples the exponent.

Exercise 5.8

Uniform daily log returns, summed over a year.

```
days, n = 252, 400_000
sigma, dt = 0.20, 1 / 252
mu_day = (r - 0.5 * sigma**2) * dt
half_width = sigma * math.sqrt(dt) * math.sqrt(3)       # uniform on [-w, w] has variance w^2
/ 3
rng = np.random.default_rng(6)
total = np.zeros(n)
for _ in range(days):
    total += mu_day + rng.uniform(-half_width, half_width, n)
ST = S0 * np.exp(total)
for q, z in ((0.01, -2.3263478740), (0.99, 2.3263478740)):
    lognormal = S0 * math.exp((r - 0.5 * sigma**2) * T + sigma * math.sqrt(T) * z)
    print(f"{q:.0%} quantile: uniform days {np.quantile(ST, q):.2f}, lognormal
{lognormal:.2f}")
```

What it prints.

```
1% quantile: uniform days 63.46, lognormal 63.43
99% quantile: uniform days 161.10, lognormal 160.84
```

Matching the variance. A uniform distribution on $[-w, w]$ has variance $w^2/3$ (compute $\int_{-w}^{w} x^2 \frac{1}{2w}\,dx = \frac{w^2}{3}$). Setting $w^2/3 = \sigma^2 \Delta t$ gives $w = \sigma\sqrt{\Delta t}\sqrt{3}$, so each day has the right variance.

What you see. Even with flat, bounded daily returns, the 1% and 99% quantiles of the year-end price are within about 0.2% of the lognormal ones: the central limit theorem at work. (The tails far beyond 99% converge more slowly; a sum of bounded steps can never exceed 252 times the bound.)

Self-test

1. The lognormal is skewed to the right: prices are bounded below by 0 but not above, and a few large rises pull the mean above the typical outcome. Formally, mean / median $= e^{\frac{1}{2}\sigma^2 T} > 1$.

2. Nothing: $E[S_T] = S_0 e^{rT} = 103.0455$ for any volatility, because the $-\frac{1}{2}\sigma^2 T$ in the exponent cancels Jensen's gap exactly. What changes is the spread, and the median, which falls.

3. $\ln E[S_T]$. The logarithm is concave, so Jensen goes the other way: $E[\ln S_T] \leq \ln E[S_T]$. Here $E[\ln(S_T/S_0)] = (r - \frac{1}{2}\sigma^2)T = 0.0100$, against $\ln E[S_T/S_0] = rT = 0.0300$.

SOLUTIONS TO LESSON 6

Convergence and Inequalities

Solutions to Exercises 6.1 to 6.8 and the Lesson 6 self-test.

Exercise 6.1

Markov. Let $Y \geq 0$ and $a > 0$.

Step 1. Outcome by outcome, $Y \geq a\,\mathbf{1}\{Y \geq a\}$:

- if $Y \geq a$, the right side is a, which is $\leq Y$;
- if $Y < a$, the right side is 0, which is $\leq Y$ because $Y \geq 0$.

Step 2. Take expectations. The expectation of an indicator is the probability of its event, so $E[Y] \geq a\,P(Y \geq a)$. Divide by $a > 0$:

$$P(Y \geq a) \leq \frac{E[Y]}{a}. \square$$

Chebyshev. Let X have mean μ and variance v, and $\varepsilon > 0$.

Step 1. $|X - \mu| \geq \varepsilon$ exactly when $(X - \mu)^2 \geq \varepsilon^2$ (both sides are non-negative, and squaring preserves order for non-negative numbers).

Step 2. Apply Markov to $Y = (X - \mu)^2 \geq 0$ with $a = \varepsilon^2$:

$$P(|X - \mu| \geq \varepsilon) = P\big((X - \mu)^2 \geq \varepsilon^2\big) \leq \frac{E[(X - \mu)^2]}{\varepsilon^2} = \frac{v}{\varepsilon^2}. \square$$

Exercise 6.2

Step 1. Scale. Write $Y_i = \sqrt{T/n}\, Z_i$ with Z_i independent standard normals. Then $Y_i^2 = \frac{T}{n} Z_i^2$.

Step 2. Facts about Z^2 (Lesson 4): $E[Z^2] = 1$ and $\mathrm{Var}(Z^2) = E[Z^4] - (E[Z^2])^2 = 3 - 1 = 2$.

Step 3. The mean. By linearity,

$$E[Q_n] = \sum_{i=1}^{n} \frac{T}{n}\, E[Z_i^2] = n \cdot \frac{T}{n} \cdot 1 = T.$$

Step 4. The variance. The constant T/n comes out of each variance squared, and because the Z_i^2 are independent their variances add:

$$\mathrm{Var}(Q_n) = \sum_{i=1}^{n} \frac{T^2}{n^2}\, \mathrm{Var}(Z_i^2) = n \cdot \frac{T^2}{n^2} \cdot 2 = \frac{2T^2}{n}. \qquad \square$$

Exercise 6.3

Step 1. $|Y_i| = \sqrt{T/n}\,|Z_i|$, because $\sqrt{T/n} > 0$.

Step 2. By Lesson 4, $E|Z| = \sqrt{2/\pi}$, so $E|Y_i| = \sqrt{T/n}\,\sqrt{2/\pi}$.

Step 3. By linearity (no independence needed),

$$E\Big[\sum_{i=1}^{n} |Y_i|\Big] = n\sqrt{\frac{T}{n}}\sqrt{\frac{2}{\pi}} = \sqrt{n^2 \cdot \frac{T}{n} \cdot \frac{2}{\pi}} = \sqrt{\frac{2nT}{\pi}}.$$

It grows like $\sqrt{n}$, without limit. For $n = 10^5$ and $T = 1$ it is 252.31.

Exercise 6.4

Write $v = \mathrm{Var}(S_T) = 20.8169^2$. The average of n independent prices has mean $E[S_T]$ and variance v/n.

Chebyshev. We want $P(|\bar{X}_n - E[S_T]| \geq 0.1) \leq 0.05$. Chebyshev bounds the left side by $\frac{v/n}{0.1^2}$, so it is enough that

$$\frac{v}{n \times 0.01} \leq 0.05 \quad \Longleftrightarrow \quad n \geq \frac{v}{0.05 \times 0.01} = 866,687.$$

Normal approximation. By the central limit theorem $\bar{X}_n$ is close to $N(E[S_T], v/n)$. A normal is within 1.96 standard deviations of its mean with probability 95%, so we need $1.96\sqrt{v/n} = 0.1$:

$$n = \Big(\frac{1.96\,\mathrm{sd}}{0.1}\Big)^2 = 166,467.$$

Compare. Chebyshev asks for 5.2 times as many paths. It is a guarantee for every distribution with this variance, so it is cautious for this particular one.

Exercise 6.5

Markov. $S_T \geq 0$ and $E[S_T] = 103.0455$, so

$$P(S_T \geq 150) \leq \frac{103.0455}{150} = 0.6870.$$

The actual probability (Lesson 5). $S_T \geq 150$ exactly when $L = \ln(S_T/S_0) \geq \ln 1.5 = 0.4055$, and $L \sim N(0.01, 0.2^2)$. Standardise:

$$P(S_T \geq 150) = P\Big(Z \geq \frac{0.4055 - 0.01}{0.2}\Big) = 1 - N(1.9773) = 0.0240.$$

Compare. Markov says "at most 69%"; the truth is about 2.4%. The bound is true but very loose, because it uses only the mean. Knowing the whole distribution gives far sharper answers; Markov is for when you don't.

Exercise 6.6

Convergence in probability. Fix $\varepsilon > 0$ (take $\varepsilon \leq n$, which holds for large n). X_n is either 0 or n, so $|X_n - 0| \geq \varepsilon$ happens only when $X_n = n$:

$$P(|X_n| \geq \varepsilon) = P(X_n = n) = \frac{1}{n} \longrightarrow 0.$$

So $X_n \to 0$ in probability.

The mean. $E[X_n] = n \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = 1$ for every n. The mean does **not** converge to the limit 0.

The variance. $E[X_n^2] = n^2 \cdot \frac{1}{n} = n$, so $\mathrm{Var}(X_n) = n - 1^2 = n - 1 \to \infty$.

What it shows. "Mean to a constant, variance to zero" is enough for convergence, but not necessary. Here X_n converges although its variance explodes: a very rare but very large value keeps the mean at 1 while becoming ever less likely. Expectations can be dominated by rare events, which is also why Monte Carlo is slow for payoffs with heavy tails.

Exercise 6.7

Estimating P(S_T > K) with a 95% interval.

```
import math

import numpy as np

S0, K, r, sigma, T = 100.0, 105.0, 0.03, 0.20, 1.0
N = lambda x: 0.5 * (1 + math.erf(x / math.sqrt(2)))
d_minus = (math.log(S0 / K) + (r - 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
rng = np.random.default_rng(11)
for n in (10**3, 10**4, 10**5, 10**6):
    ST = S0 * np.exp((r - 0.5 * sigma**2) * T + sigma * math.sqrt(T) *
rng.standard_normal(n))
    p = (ST > K).mean()
    half = 1.96 * math.sqrt(p * (1 - p) / n)
    print(f"n = {n:>7}: {p:.4f} +/- {half:.4f}")
print(f"N(d-) = {N(d_minus):.4f}")
```

What it prints.

```
n =    1000: 0.4450 +/- 0.0308
n =   10000: 0.4279 +/- 0.0097
n =  100000: 0.4218 +/- 0.0031
n = 1000000: 0.4232 +/- 0.0010
N(d-) = 0.4231
```

Why this interval. Each estimate is an average of the indicator $\mathbf{1}\{S_T > K\}$: a random variable that is 1 with probability p and 0 otherwise. Its variance is $p(1-p)$ (Lesson 4's coin: $E[X^2] - E[X]^2 = p - p^2$). So the average has standard deviation $\sqrt{p(1-p)/n}$, and the 95% half-width is $1.96\sqrt{p(1-p)/n}$.

What you see. The half-width shrinks tenfold for every hundredfold increase in paths, the $\sqrt{n}$ law. Every interval contains the exact $N(d_-) = 0.4231$; at a million paths the half-width is about 0.0010.

Exercise 6.8

Squared and absolute steps, up to 100,000 of them.

```
T = 1.0
rng = np.random.default_rng(8)
for n in (10, 1000, 100_000):
    Y = rng.normal(0, math.sqrt(T / n), (200, n))
    Q, A = (Y**2).sum(axis=1), np.abs(Y).sum(axis=1)
    print(f"n = {n:>6}: Q mean {Q.mean():.4f}, sd {Q.std():.4f} (theory {math.sqrt(2 / n) *
T:.4f}); "
          f"sum |Y| mean {A.mean():.2f} (theory {math.sqrt(2 * n * T / math.pi):.2f})")
```

What it prints.

```
n =     10: Q mean 1.0314, sd 0.4416 (theory 0.4472); sum |Y| mean 2.56 (theory 2.52)
n =   1000: Q mean 1.0003, sd 0.0483 (theory 0.0447); sum |Y| mean 25.23 (theory 25.23)
n = 100000: Q mean 0.9999, sd 0.0043 (theory 0.0045); sum |Y| mean 252.28 (theory 252.31)
```

What you see. The sum of squares stays centred on $T = 1$ while its standard deviation shrinks like $\sqrt{2/n}$ (to 0.0045 at $n = 10^5$); the sum of absolute values grows like $\sqrt{2n/\pi}$. Both match Exercises 6.2 and 6.3.

Self-test

1. No. Chebyshev needs only a finite variance; that is exactly why it holds for every distribution, and why it is loose for any particular one, such as the normal.

2. To 2. The mean $2 + 1/n \to 2$ and the variance $1/n \to 0$, so by "mean to a constant, variance to zero", $X_n \to 2$ in probability (and in mean square).

3. Each absolute step is of size $\sqrt{T/n}$, and there are n of them: $n\sqrt{T/n} = \sqrt{nT} \to \infty$. Each squared step is of size T/n, and n of them add up to T, with fluctuations that shrink like $1/\sqrt{n}$. A path made of random steps of size $\sqrt{\Delta t}$ has infinite length but finite quadratic variation.